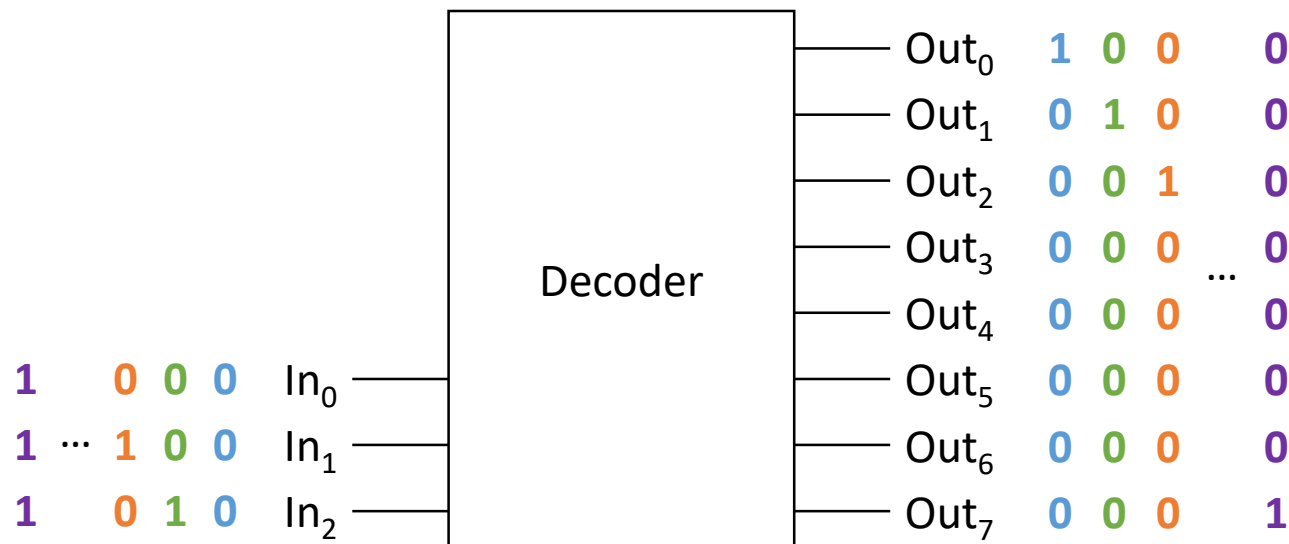


Lecture 6: Combinational Logic Blocks

CMPS 221 – Computer Organization and Design

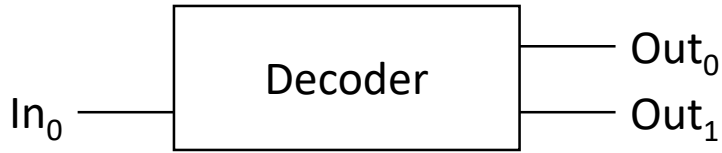
Decoders

- A **decoder** is a logic block with n input signals and 2^n output signals, where only one output signal is **1** for every input combination
 - e.g., 3-to-8 decoder

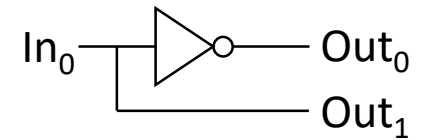


Implementing Decoders

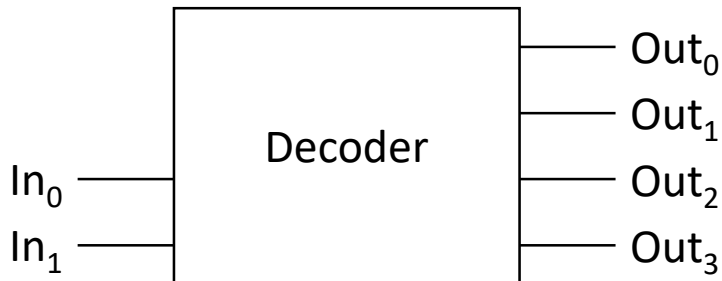
1-to-2 decoder:



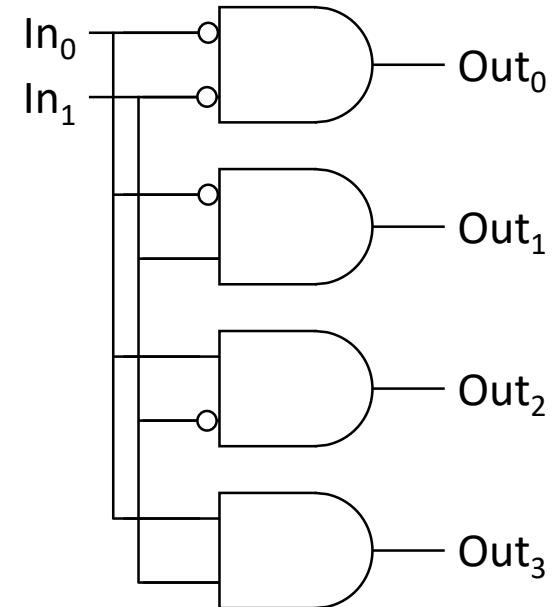
Inputs	Outputs	
In_0	Out_0	Out_1
0	1	0
1	0	1



2-to-4 decoder:

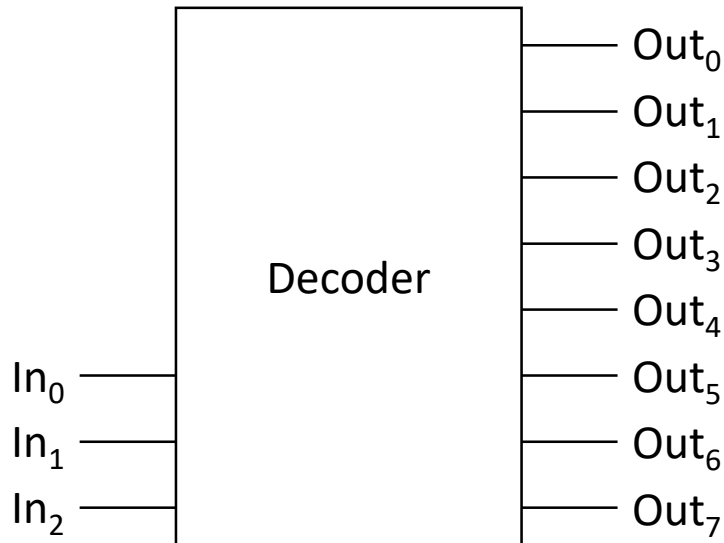


Inputs		Outputs			
In_0	In_1	Out_0	Out_1	Out_2	Out_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

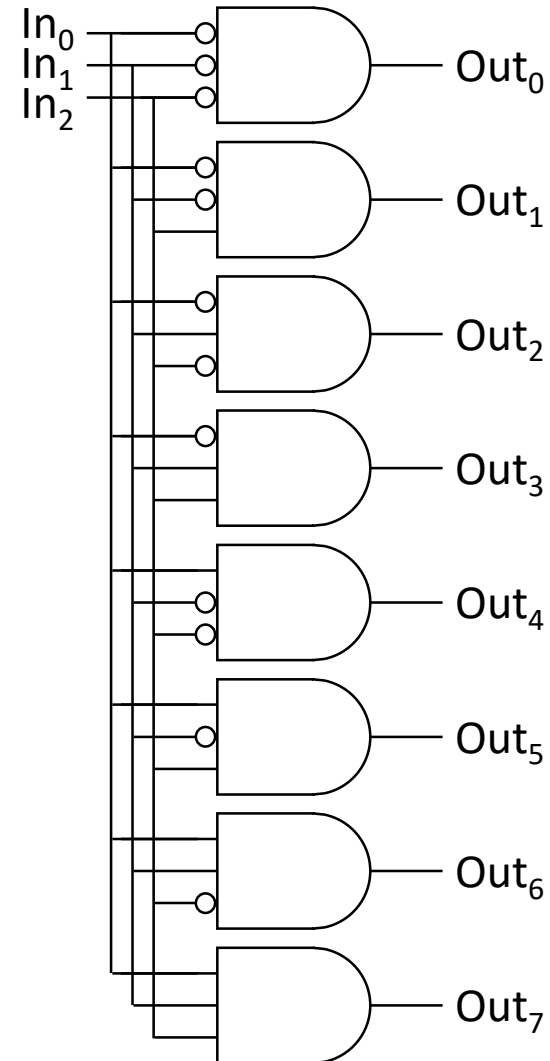


Implementing Decoders

3-to-8 decoder:

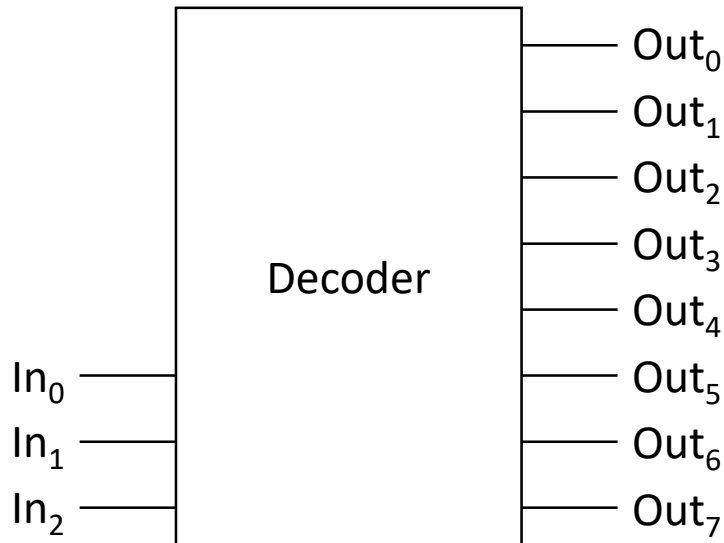


Inputs			Outputs							
In_0	In_1	In_2	Out_0	Out_1	Out_2	Out_3	Out_4	Out_5	Out_6	Out_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



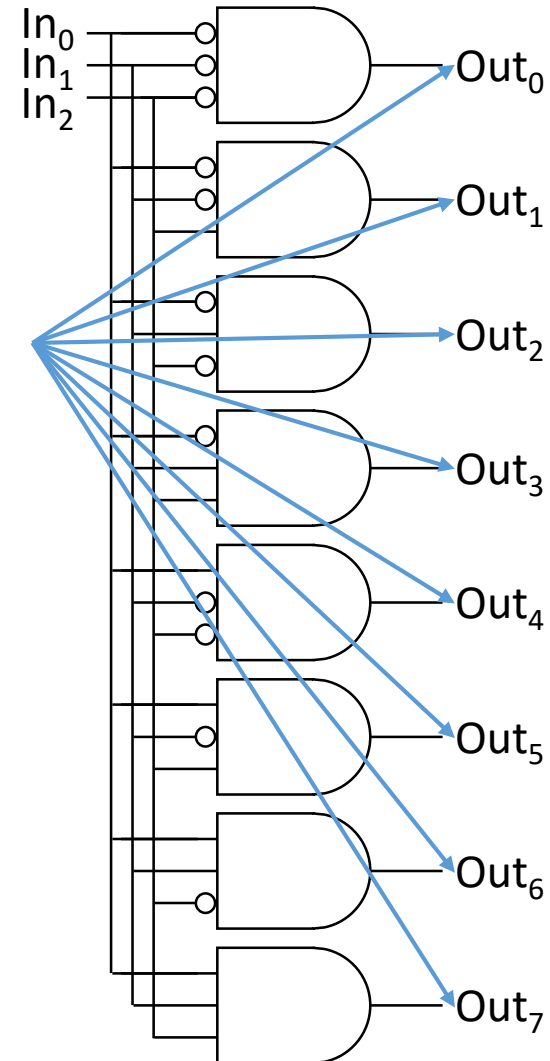
Implementing Decoders

3-to-8 decoder:



Every output corresponds
to a minterm

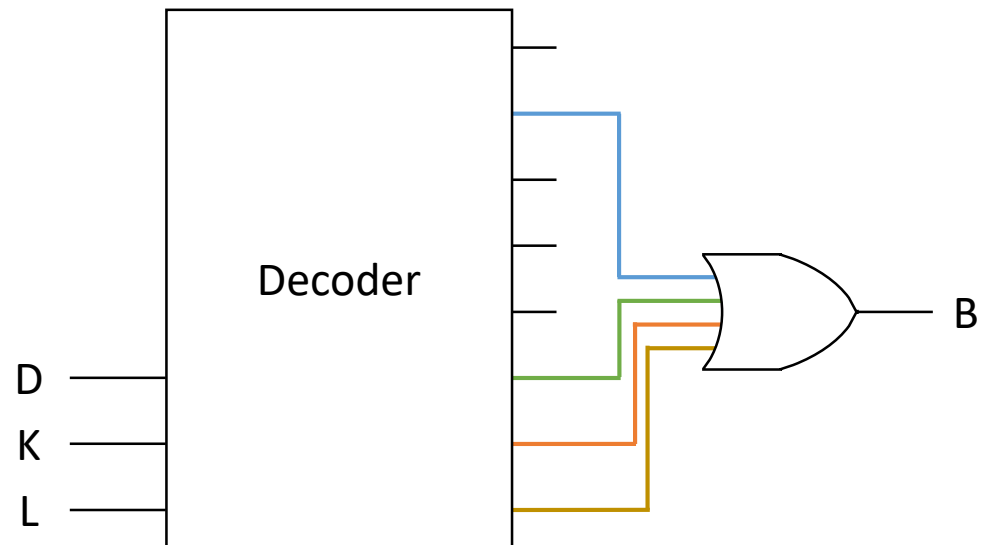
Inputs			Outputs							
In_0	In_1	In_2	Out_0	Out_1	Out_2	Out_3	Out_4	Out_5	Out_6	Out_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1



Logic Blocks with Decoders

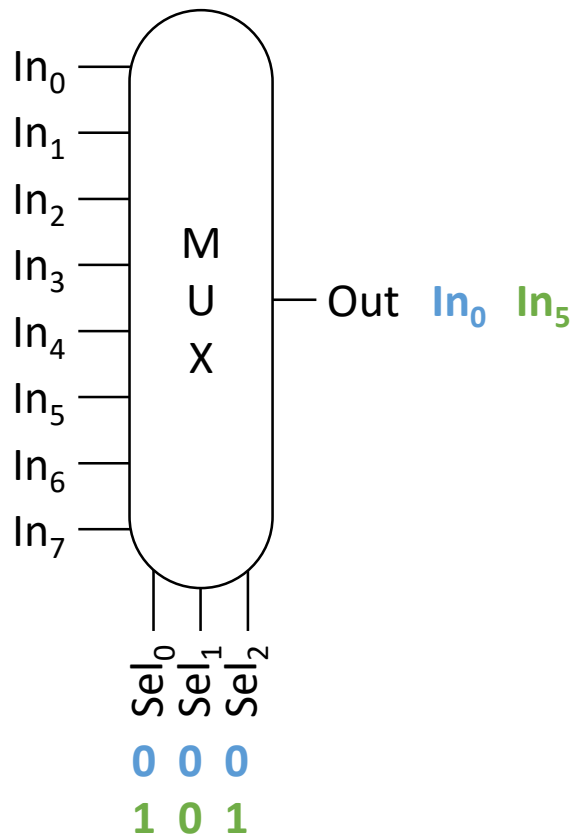
- Since every output in a decoder corresponds to a minterm, decoders can also be cascaded with OR gates implement any logic block

Inputs			Output
D	K	L	B
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



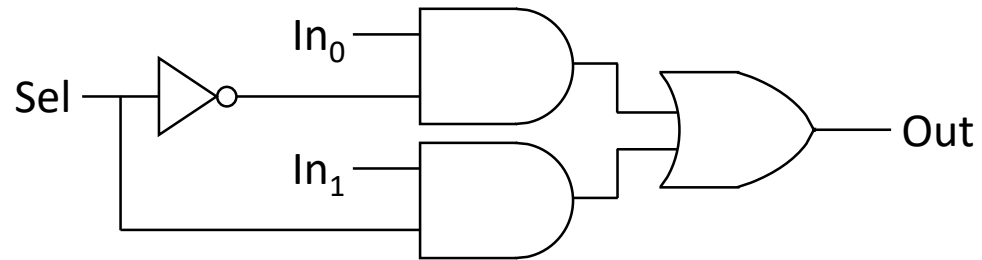
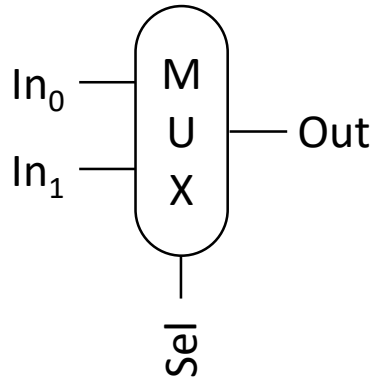
Multiplexers

- A **multiplexer** (or **mux**) is a logic block that selects one of multiple input signals based on a selector signal

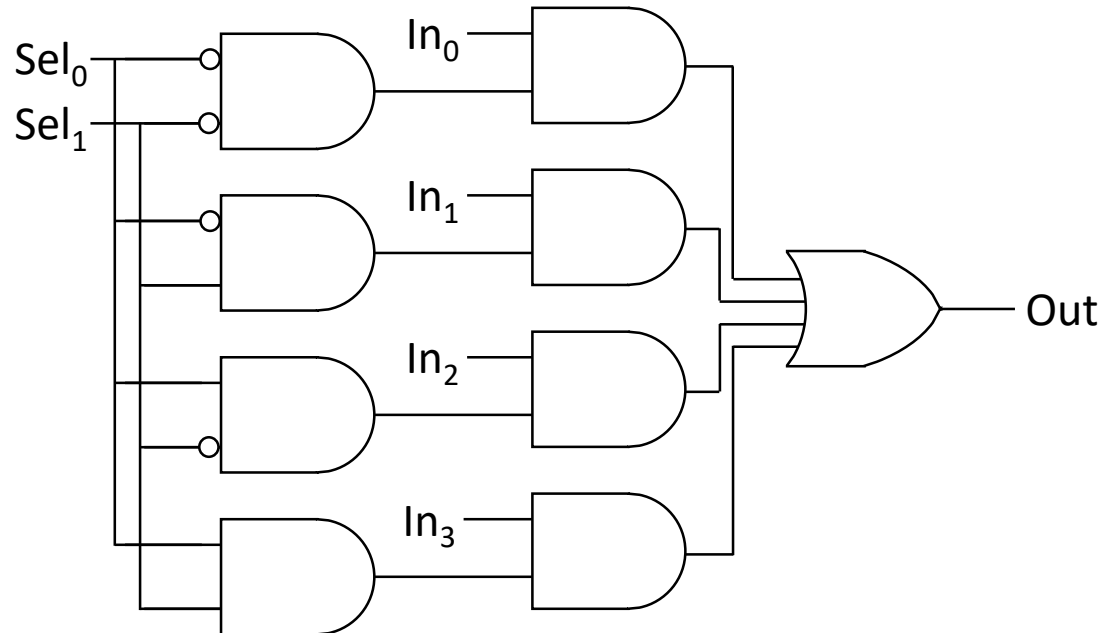
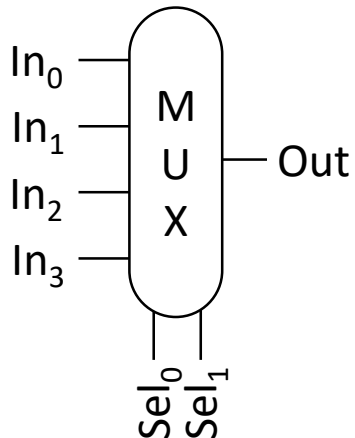


Implementing Multiplexers

2-input mux:

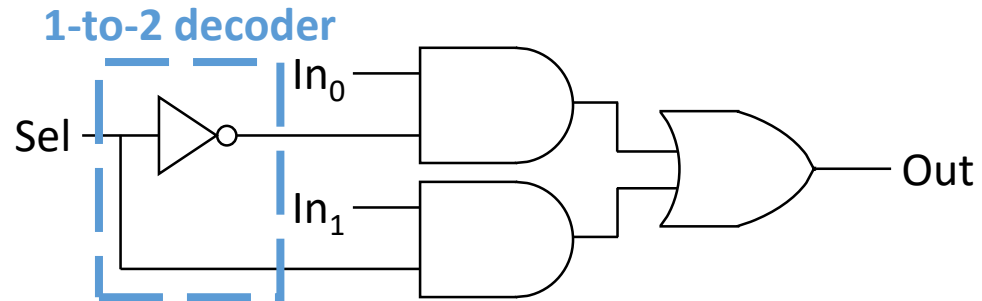
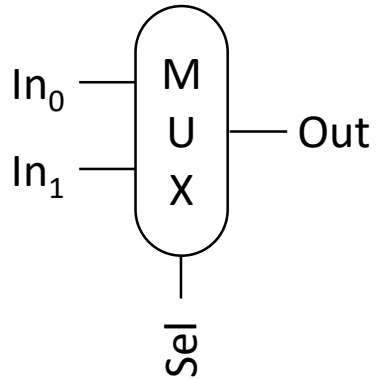


4-input mux:

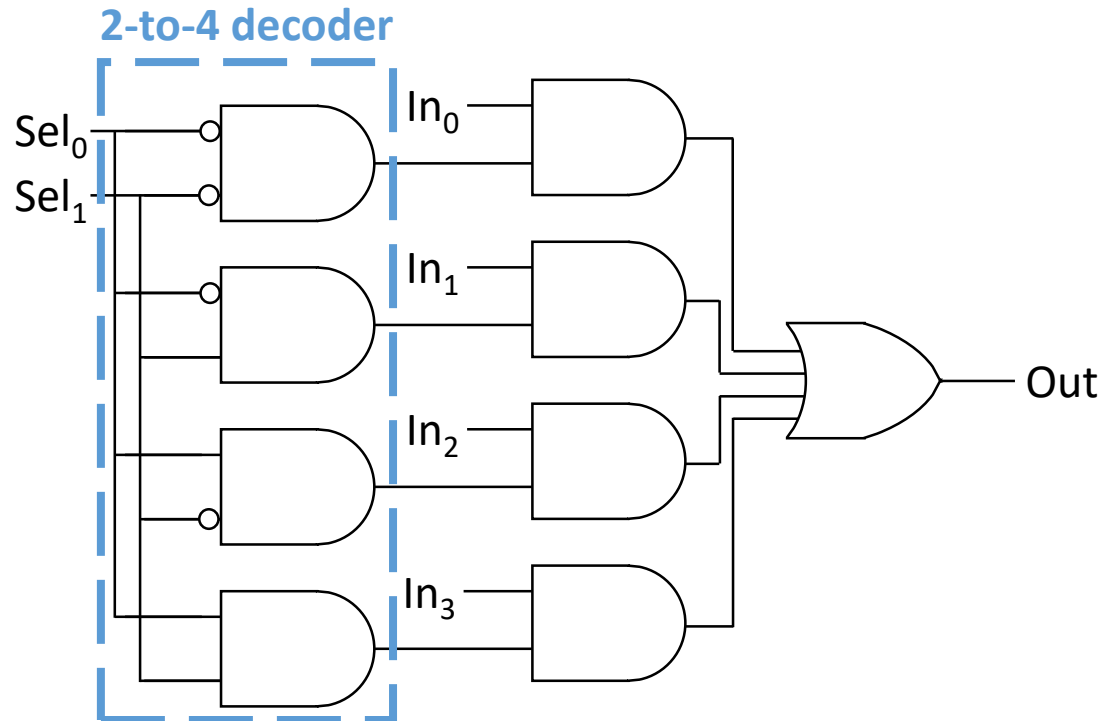
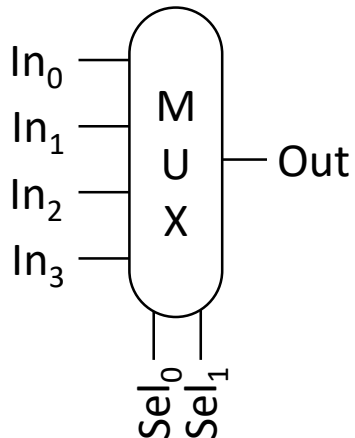


Implementing Multiplexers

2-input mux:

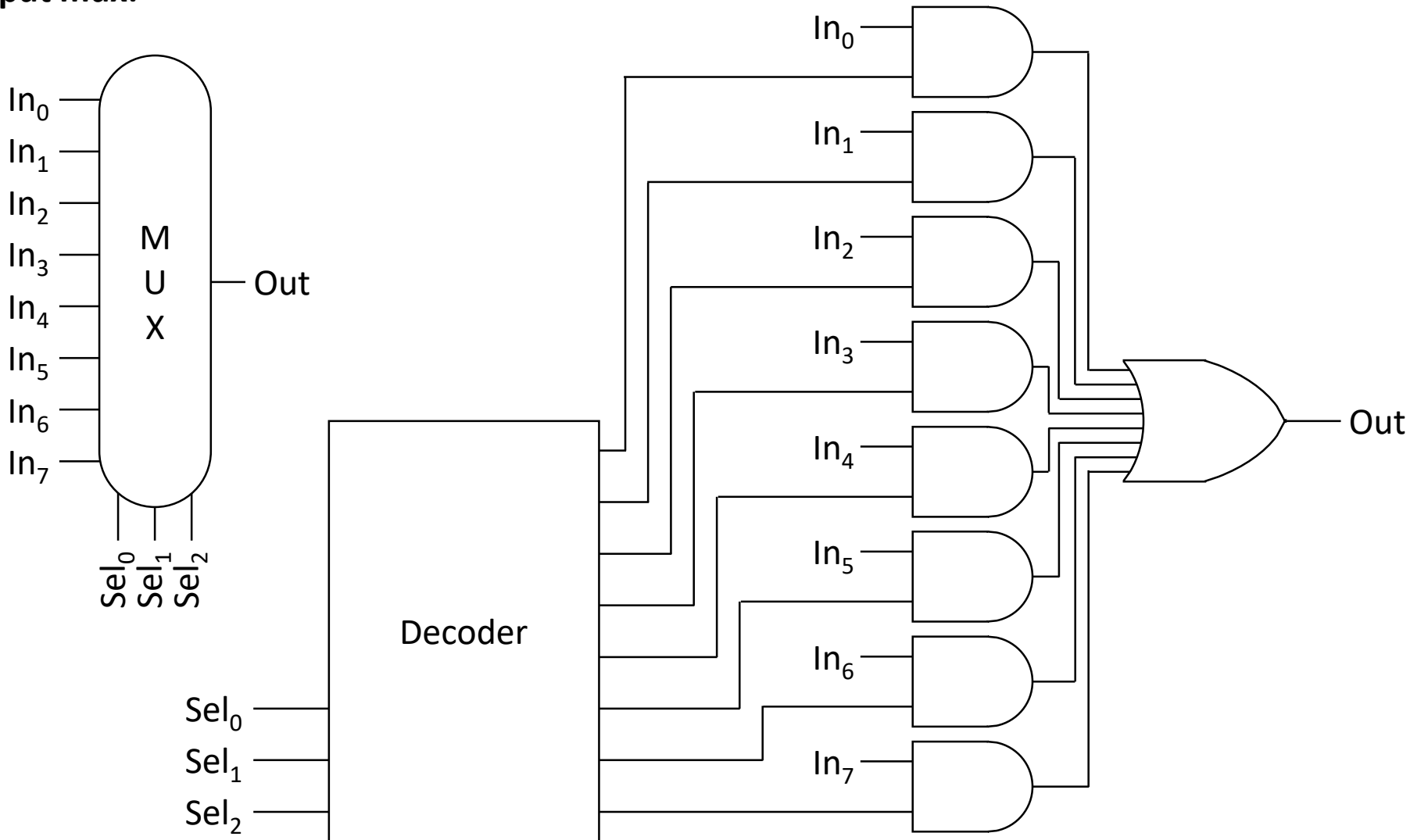


4-input mux:



Implementing Multiplexers

8-input mux:



Logic Blocks with Multiplexers

- Multiplexers can also be used to implement any logic block by using all but one input as the selector signal and expressing the final output in terms of the remaining signal

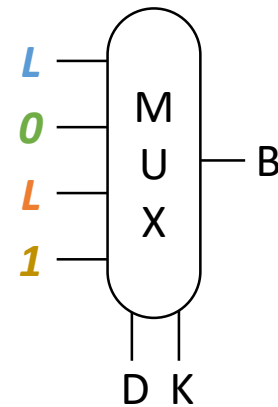
Inputs			Output
D	K	L	B
0	0	0	0
		1	1
0	1	0	0
		1	0
1	0	0	0
		1	1
1	1	0	1
		1	1

$$B = L$$

$$B = 0$$

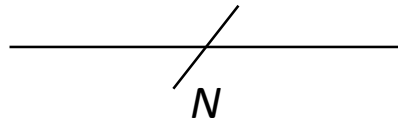
$$B = L$$

$$B = 1$$

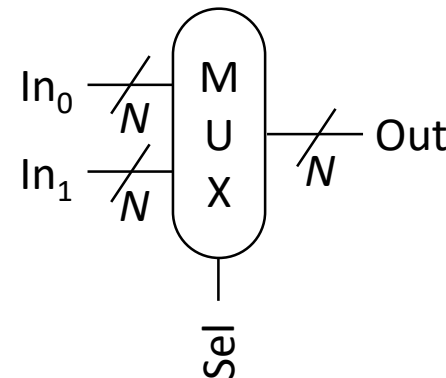


Multi-bit Multiplexers

- Operations are often performed on a collection of wires (multi-bit signals), not just a single one
- A **bus** is a collection of wires treated as one signal
- In circuit diagrams, a N -bit bus is drawn as follows:

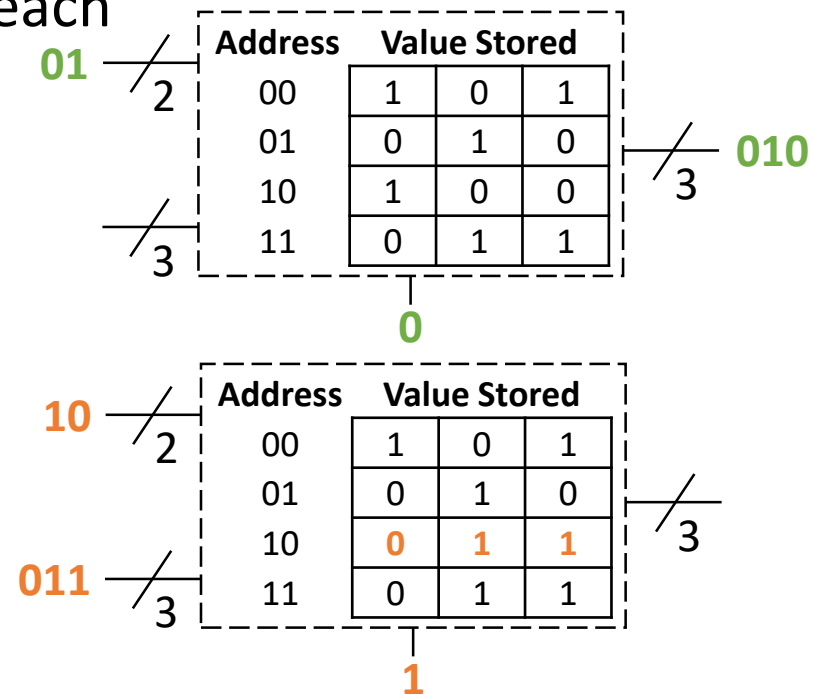
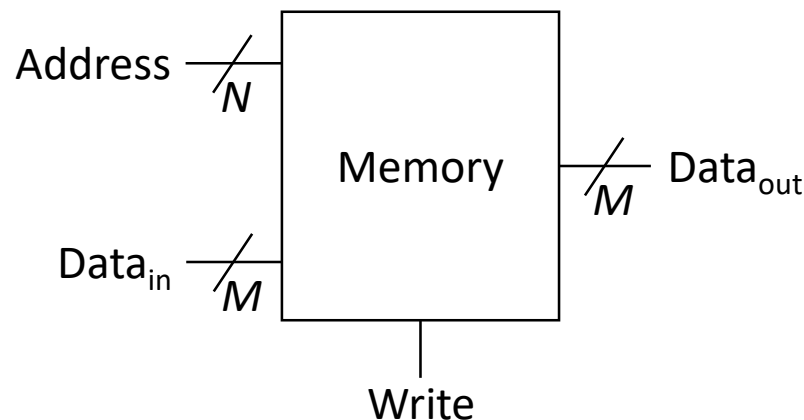


- We often need to select between multi-bit signals
 - Draw a single mux with buses as inputs/outputs, instead of multiple mux's
 - Decoder is shared across wires in the bus



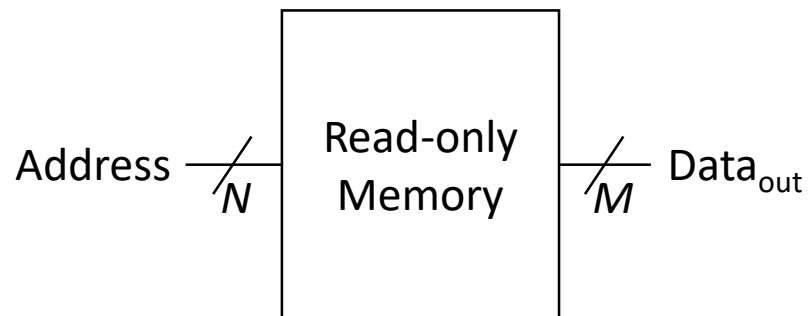
Memory

- A **memory** is a logic block that stores data
 - It takes a memory address as input and outputs the value stored at that address or writes to it
 - A memory with a N -bit address and M -bit output value stores 2^N entries with M bits each
 - Different technologies for building memories will be covered later



Read-only Memory

- A **read-only memory (ROM)** is a type of memory that is:
 - Read-only
 - Its contents can only be read, and cannot be modified after it is manufactured
 - Nonvolatile
 - Its contents are not erased when the power is turned off



Logic Blocks with ROMs

- ROMs can also be used to implement any logic block by simply storing the truth table as the content of the memory

Inputs			Output	
A	B	C	D	E
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	1

Address	Values Stored	
000	0	0
001	0	1
010	1	1
011	1	0
100	1	0
101	0	0
110	1	1
111	1	1

Textbook Sections

- Some of the content in these slides corresponds to:
 - Textbook:
 - *Computer Organization and Design, 5th Edition by David Patterson and John Hennessy, Morgan Kaufmann, 2014.*
 - Sections:
 - B.3